

Efficient Long Sequence Modeling

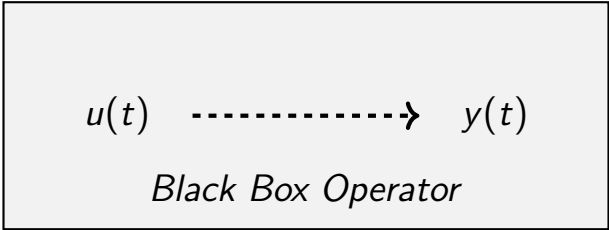
From LSSL to S4: Solving the Kernel Bottleneck

Deep Dive: DPLR & Woodbury

January 22, 2026

1. Problem: Sequence-to-Sequence Operator

$$\mathcal{F} : u \mapsto y$$

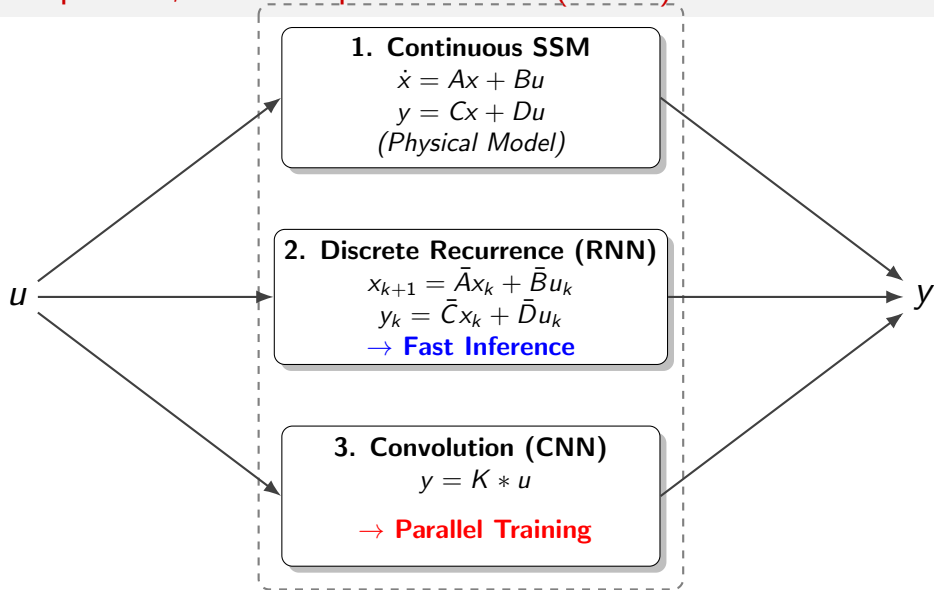


A diagram illustrating a black box operator. It consists of a light gray rectangular box with a black border. Inside the box, the input $u(t)$ is on the left, followed by a dashed arrow pointing to the right, and the output $y(t)$ is on the right. Below the arrow, the text "Black Box Operator" is written in an italicized font.

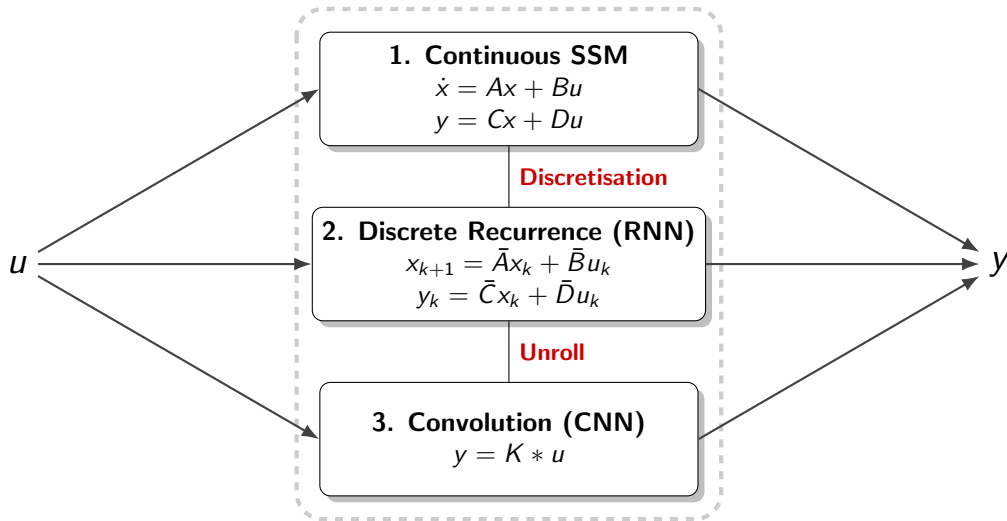
$$u(t) \quad \text{-----} \rightarrow \quad y(t)$$

Black Box Operator

2. Same Operator, Three Representations (Views)



3. Between views



4. Derivation between views

Step 1: Discretization

Transform $(A, B) \rightarrow (\bar{A}, \bar{B})$ via Bilinear:

$$\bar{A} = (I - \frac{\Delta}{2}A)^{-1}(I + \frac{\Delta}{2}A)$$

$$\bar{B} = (I - \frac{\Delta}{2}A)^{-1}\Delta B$$

$$\bar{C} = C$$

$$\bar{D} = D$$

Step 2: Unroll Recurrence

$$x_0 = 0$$

$$y_0 = \bar{C}x_0 = 0$$

$$x_1 = \bar{B}u_0$$

$$y_1 = \bar{C}x_1 = \bar{C}\bar{B}u_0 + \dots$$

$$x_2 = \bar{A}x_1 + \bar{B}u_1 = \bar{A}\bar{B}u_0 + \bar{B}u_1$$

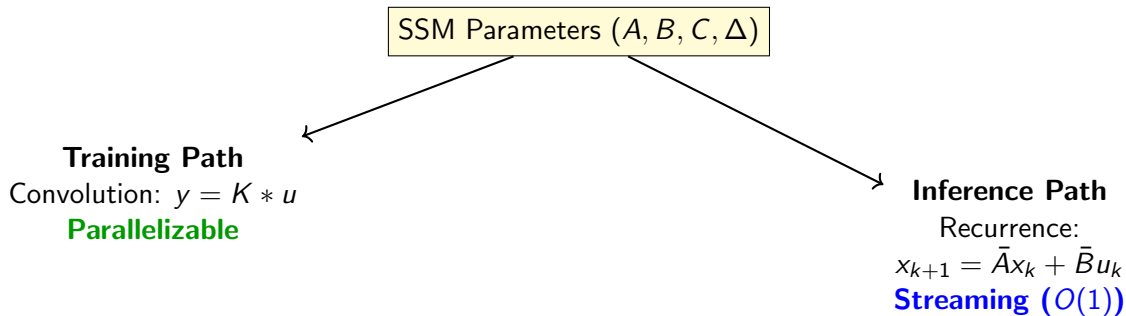
$$y_2 = \bar{C}x_2 = \bar{C}\bar{A}\bar{B}u_0 + \bar{C}\bar{B}u_1 + \dots$$

Identify Convolution Kernel:

$$K = \{\underbrace{\bar{C}\bar{B}}_{K_0}, \underbrace{\bar{C}\bar{A}\bar{B}}_{K_1}, \underbrace{\bar{C}\bar{A}^2\bar{B}}_{K_2}, \dots\}$$

$$y_k = \sum_{t=0}^{k-1} K_t u_{k-1-t}$$

5. Switch between views



6. Reality Check: Why is it still slow?

1. From Recurrence to Convolution

$$x_{k+1} = \bar{A}x_k + \bar{B}u_k$$

$$y_k = \bar{C}x_k \quad (\text{Assume } x_0 = 0)$$

Expand recurrence (History accumulation):

$$x_k = \sum_{i=0}^{k-1} \bar{A}^{k-1-i} \bar{B}u_i$$

Substitute into y_k and change var to **lag** $t = k - 1 - i$:

$$y_k = \sum_{t=0}^{k-1} \underbrace{(\bar{C}\bar{A}^t\bar{B})}_{K_t} u_{k-1-t}$$

2. The Computational Cost

To use the convolution view, we must materialize K :

$$K = [\bar{C}\bar{B}, \bar{C}\bar{A}\bar{B}, \dots, \bar{C}\bar{A}^{L-1}\bar{B}]$$

- **Long Kernel:** Sequence length L is large (e.g., 10k+).
- **Cost:** If $\bar{A}$ is dense ($N \times N$):
 - Each step: $O(N^2)$.
 - Total for L taps: $O(LN^2)$.

7. Reframe: Kernel as a Matrix Function

We switch from **Time Domain** to **Generating Functions** (Z-transform).

Step 1: Define the Generating Function

$$\mathcal{K}(z) = \sum_{t=0}^{\infty} K_t z^t = \sum_{t=0}^{\infty} (\bar{C} \bar{A}^t \bar{B}) z^t$$

Step 2: Factor out constant matrices

$$\mathcal{K}(z) = \bar{C} \left(\sum_{t=0}^{\infty} (z \bar{A})^t \right) \bar{B}$$

Step 3: Apply Matrix Geometric Series (*ln small $|z|$*)

$$\sum_{t=0}^{\infty} (z \bar{A})^t = (I - z \bar{A})^{-1}$$

8. Bridge: Bilinear Resolvent

Step 0: What we want (Discrete Resolvent)

$$(I - z\bar{A})^{-1}$$

Step 1: Bilinear Discretization

$$\bar{A} = (I - \frac{\Delta}{2}A)^{-1}(I + \frac{\Delta}{2}A), \quad \bar{B} = (I - \frac{\Delta}{2}A)^{-1}\Delta B$$

Step 2: Plug $\bar{A}$ into resolvent $\rightarrow$ Simplify (Key Identity derived in Appendix B)

$$\boxed{(I - z\bar{A})^{-1} = (I - \frac{\Delta}{2}A)(\alpha(z)I - A)^{-1}}$$

$$\alpha(z) = \frac{2}{\Delta} \frac{1-z}{1+z}$$

Note: Let $\zeta = \alpha(z)$. Below we write the resolvent as $(\zeta I - A)^{-1}$. (We rename $\zeta \rightarrow z$ for simplicity).

9. Matrix inversion

We need many resolvent inversions:

$$(zI - A)^{-1} \quad \text{for many } z$$

- **Dense inverse per z :** $O(N^3)$ (Gaussian elimination).
- **Many z points:** Long sequence $L \implies$ many evaluations $\implies$ Cost explodes.

Complexity

Bottleneck = Resolvent Inversion.

10. Fix: DPLR + Woodbury

(A) DPLR Structure

$$A = \Lambda + PQ^\top \quad (\Lambda \text{ diagonal, } P, Q \in \mathbb{R}^{N \times r})$$

(B) Woodbury Identity (r=1)

$$(\Lambda + PQ^\top)^{-1} = \Lambda^{-1} - \frac{\Lambda^{-1}PQ^\top\Lambda^{-1}}{1 + Q^\top\Lambda^{-1}P}$$

When **A** is Hippo Leg-S, what we need $(zI - A)^{-1}$

Reduced to the form of $(\Lambda + PQ^\top)^{-1}$

Complexity

Per z : **$O(N)$ instead of $O(N^3)$.**

11. Big Picture: How the problem evolves

